

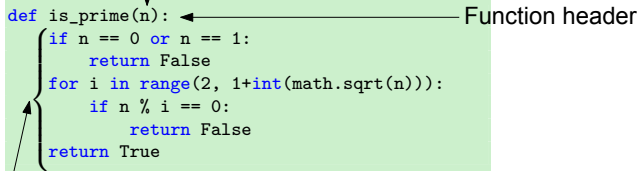
Python: functions

Intro. to computing

`https://mandarm.github.io/computing.html`

Functions

Function parameters (zero or more, comma separated)



```
def is_prime(n):  
    if n == 0 or n == 1:  
        return False  
    for i in range(2, 1+int(math.sqrt(n))):  
        if n % i == 0:  
            return False  
    return True
```

The diagram shows a Python function definition for `is_prime(n)`. An arrow points from the text "Function parameters (zero or more, comma separated)" to the parameter `n` in the function signature. Another arrow points from the text "Function header" to the `def is_prime(n):` line. A third arrow points from the text "Function body" to the indented lines of the function. The function body contains an `if` statement, a `for` loop with an `if` statement inside, and a final `return True` statement.

Function body

■ `return` statements

- multiple occurrences permitted (as in example above)
- may be followed by an optional expression (including `None`)

■ *Parameters* (or formal parameters): `n` in the above example

■ *Arguments* (or actual parameters): value passed when the function is actually called

Example: `is_prime(4)`, `is_prime(5)` — 4 and 5 are arguments